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Development of estimation algorithms for navigation of unmanned aerial vehicles

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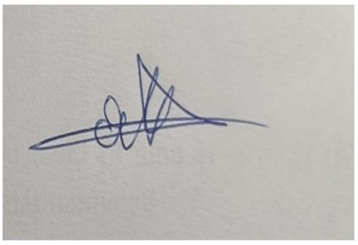

Abstract

Accurate attitude estimation and prediction are critical for the autonomous navigation and advanced control of unmanned aerial vehicles (UAVs). While classical sensor fusion algorithms effectively estimate current vehicle orientation, they are fundamentally reactive and cannot anticipate future states. Modern control strategies, such as Model Predictive Control (MPC), require multi-step ahead predictions of future attitude, creating a need for predictive models based on deep learning. This thesis develops a comprehensive framework for UAV attitude prediction using Long Short-Term Memory (LSTM) network and attention-based mechanism are explored as an extension. The proposed architecture is designed to predict future roll ϕ, pitch θ, and yaw ψ angles over a configurable multi-step horizon using look-back windows of IMU measurements and control inputs. To establish a robust training dataset and baseline performance, four classical tilt estimation filters—the Complementary Filter, Mahony Filter, Explicit Complementary Filter, and Extended Kalman Filter (EKF)—were implemented and optimized using real flight data. By utilizing the EKF-estimated angles as training labels, the framework becomes self-contained and applicable to real-world scenarios where external ground-truth systems are unavailable. Experimental evaluation demonstrates that LSTM model not only achieves high estimation accuracy but also provides the essential multi-step prediction capability required for proactive MPC-based flight controllers, effectively bridging the gap between classical sensor fusion and modern deep learning for UAV navigation.

Keywords

UAVs, Complementary Filtering, EKF, Attitude estimation, Classic estimation methods, Neural Networks, Machine Learning, MPC, LSTM, RNN, Sensor Fusion

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Acronym Index

CNN: Convolutional Neural Network
DOF: Degrees of Freedom
EKF: Extended Kalman Filter
GNSS: Global Navigation Satellite System
GRU: Gated Recurrent Unit
IMU: Inertial Measurement Unit
LSTM: Long Short-Term Memory
MAE: Mean Absolute Error
MPC: Model Predictive Control
MSE: Mean Squared Error
PID: Proportional-Integral-Derivative
RMSE: Root Mean Square Error
RNN: Recurrent Neural Network
SO(3): Special Orthogonal Group in 3 Dimensions
SVD: Singular Value Decomposition
UAV: Unmanned Aerial Vehicle

INTRODUCTION

Unmanned Aerial Vehicles (UAVs) have emerged as transformative technologies across many sectors, from aerial photography to package delivery. At the heart of every autonomous UAV lies a critical capability: knowing its own attitude (orientation) in three-dimensional space. Without continuous and accurate attitude information, a UAV cannot maintain stable flight or respond to disturbances.

Modern UAVs rely on Inertial Measurement Units (IMUs), combining gyroscopes and accelerometers. Because individual sensors have inherent limitations, classical sensor fusion algorithms like the Extended Kalman Filter (EKF) are used to produce reliable estimates of the current tilt angles. However, while effective for real-time estimation, these approaches are fundamentally reactive—they estimate the present state but cannot anticipate the future.

Modern advanced flight control strategies, such as Model Predictive Control (MPC), require multi-step ahead predictions of future states over a finite horizon [12]. Recent advances in deep learning, particularly Long Short-Term Memory (LSTM) networks, have shown strong capabilities in learning temporal dependencies [9]. Yet, the majority of current neural network-based approaches in UAV navigation focus solely on single-step estimation [10][11]. There is a critical need for predictive models that provide the multi-step prediction horizons required by MPC-based flight controllers.

The subject of this thesis

This thesis addresses the problem of predicting future UAV attitude states from sequential sensor and control data. Specifically, given a look-back window of timesteps, the task is to predict the attitude angles for the next timesteps into the future.

Because true attitude angles are not directly measurable by a single sensor, this creates a two-stage problem:

1. **Estimation:** Accurate attitude angles must first be produced from raw IMU measurements.
2. **Prediction:** These estimated angles must then be used to train a predictive model capable of forecasting future states.

The quality of the first stage directly constrains the second. Therefore, this thesis bridges both stages into a single integrated pipeline, starting with classical tilt estimation and ending with a deep learning predictive model.

Aim and objectives

The primary aim of this thesis is to develop a complete estimation-to-prediction pipeline for UAV attitude. The specific objectives are:

1. Implement and optimize four classic tilt estimation filters to produce accurate attitude angle estimates from raw 6-DOF IMU data, establishing a baseline.

2. Design and train an LSTM -based architecture, with an attention mechanism evaluated as an extension that predicts the attitude angles () over a future horizon of timesteps from look-back windows of sensor and control data.
3. Evaluate prediction accuracy per step and per angle using RMSE and MAE metrics.
4. Implement the complete pipeline in C++ using Eigen and LibTorch, demonstrating feasibility for real-time embedded deployment.

Proposed Methodology: LSTM-Based Predictive Framework

To execute the two-stage pipeline, the methodology is structured as follows:

First, raw 6-DOF IMU measurements are processed through an Extended Kalman Filter (EKF). The EKF provides optimal probabilistic state estimates, which serve as the baseline features and training labels for the neural network.

Second, the predictive deep learning model—an attention-based LSTM—processes a continuous look-back window of recent flight data. The input sequence consists of three distinct feature groups: the current EKF-estimated attitude angles (spatial state), raw gyroscope angular velocities (rate of change), and active PID control torques (forcing inputs).

Unlike standard recurrent architectures that compress historical data into a single hidden state, a custom attention mechanism dynamically evaluates and weights the importance of each past timestep. This allows the network to focus on relevant dynamic events like sudden control inputs or rapid maneuvers. Finally, a Direct Multi-Output regression generates the multi-step ahead forecast for roll, pitch, and yaw, fulfilling the prerequisites for proactive MPC algorithms.

Innovation

The core innovations of this work are:

- **Multi-Step vs. Single-Step:** While existing literature focuses on single-step attitude estimation to replace classical filters, this work specifically targets multi-step ahead predictions to enable proactive, rather than reactive, MPC flight control.
- **Self-Contained Training:** The LSTM is trained entirely on filter-estimated attitude angles rather than externally provided ground truth. This makes the approach applicable to real-world scenarios where external motion-capture systems are unavailable.
- **Embedded Readiness:** By implementing the network entirely in C++ using LibTorch and Eigen, the framework is immediately deployable on resource-constrained embedded systems, addressing a practical gap left by Python-heavy research models.

Structure

The remainder of this thesis is organized as follows: **Chapter 1** establishes the theoretical background, introducing the fundamentals of UAV attitude representation, inertial sensors, and the principles of sequential learning using recurrent neural networks. **Chapter 2** presents the implementation, parameter optimization, and comparative analysis of four classical tilt

estimation filters (Complementary, Mahony, Explicit Complementary, and EKF) using real flight data, establishing the data generation pipeline. **Chapter 3** introduces the core innovation of this work: the design, training, and evaluation of the attention-based LSTM architecture for multi-step ahead attitude prediction. Finally, conclusions and directions for future work and research are discussed at the end of the document

CHAPTER 1: Theoretical background

This chapter establishes the theoretical foundation for UAV attitude estimation. It begins by outlining the role of attitude in flight dynamics and detailing the inertial sensors utilized for motion measurement. Subsequently, various mathematical representations of spatial orientation are examined. The fundamental concepts of sensor fusion are then introduced, providing the basis for the classical filtering algorithms developed in Chapter 2. Finally, the chapter presents the principles of sequential learning using neural networks, establishing the theoretical framework for the deep learning approach proposed in Chapter 3.

1.1 The role of attitude in UAV flight

The autonomous operation of drones depends critically on their ability to determine and control their orientation in three-dimensional space

The attitude of a UAV describes its rotation orientation relative to a fixed reference frame, typically defined with respect to the Earth's surface. In aviation, attitude is conventionally expressed using three angular coordinates known as Euler angles:

1. Roll (ϕ): Rotation around the longitudinal axis (forward-backward direction)
2. Pitch (θ): Rotation around the lateral axis (left-right direction)
3. Yaw (ψ): Rotation around the vertical axis (up-down direction)

These angles directly influence the vehicle's ability to maintain stable flight, execute precise maneuvers, and respond appropriately to environmental disturbances such as wind gusts. Without accurate attitude information, a UAV cannot maintain level flight, execute controlled turns, provide a stable platform for cameras, or implement autonomous navigation algorithms.

Unlike position (which can be directly measured by GPS) or velocity (which can be derived from GPS updates), attitude cannot be directly measured by a single sensor. Instead, it must be estimated by fusing measurements from multiple sensors, each with different characteristics and limitations.

1.2 IMU and Sensors

An Inertial Measurement Unit (IMU) is a compact electronic device that combines multiple motion sensors to measure a vehicle's kinematic state. A typical 6-axis IMU contains an 3-axis gyroscope and 3-axis accelerometer.

1.2.1 Gyroscope

A gyroscope is an angular rate sensor that measures how fast an object rotates around one or more axes, and in UAVs it is one of core sensors for attitude estimations. In practice gyroscopes output angular velocity (e.g. in degrees or rad per second). Angular velocity is measured as $\omega \in R^3$ (rad/s). Each axis output corresponds to the rate of rotation about the x, y, z axes of the sensor frame.

Gyroscopes have key advantages and limitations. They provide very smooth short-term attitude information and are insensitive to linear accelerations, which makes them reliable during aggressive maneuvers where accelerometers are heavily corrupted by non-gravitational forces. However, their measurements contain bias and noise, so integration causes drift that grows over time, meaning gyro-only attitude estimates must be regularly corrected using other sensors such as accelerometers, magnetometers, or GNSS within a sensor fusion algorithm. [1]

1.2.2 Accelerometer

Accelerometers are fundamentally different from what their name suggests. They do not measure kinematic acceleration. Instead, they measure specific force, also called proper acceleration. It is measured as $\alpha \in R^3$ (m/s^2). Each axis output corresponds to the specific force experienced along the x, y, z axes of the sensor frame, including gravity during static conditions.

Accelerometers excel at providing long-term absolute reference through the gravity vector, allowing computation of roll and pitch via inverse trigonometry when the vehicle is mostly stationary or slow moving. Accelerometer measures $a = [ax, ay, az]$ and we can calculate roll and pitch via these equations

$$\phi = atan2(ay, az)$$

$$\theta = arsin\left(\frac{ax}{g}\right)$$

1.3 Attitude Representation

Representing three-dimensional rotations mathematically is a non-trivial problem in attitude estimation. While the physical concept of orientation is intuitive, capturing it numerically requires careful consideration of mathematical properties, computational efficiency, and singularity avoidance. This section introduces three common representations used in attitude estimation algorithms, each with distinct advantages and limitations that influence their suitability for different filtering approaches.

1.3.1 Euler Angles

Euler angles are the most intuitive representation of orientation, describing attitude as a sequence of three rotations around specified axes. The aerospace convention, known as the ZYX (yaw-pitch-roll) sequence, applies rotations in the following order [2]:

1. Rotate by ψ (yaw) around the Z-axis (vertical)
2. Rotate by θ (pitch) around the new Y-axis

3. Rotate by ϕ (roll) around the new X-axis

This three-parameter representation is highly interpretable and maintains a minimal memory footprint for computational operations

Euler angles suffer from non-uniqueness (multiple angle combinations can represent the same physical orientation) and require complex trigonometric transformations when computing angular velocity from angle rates

The Complementary Filter (Chapter 2) operates directly on roll and pitch Euler angles, taking advantage of their simplicity for the basic weighted fusion approach. Furthermore, all filters in this thesis output their final estimates as Euler angles for interpretability and comparison with ground truth measurements. For UAV flight control applications where extreme pitch angles (near ±90°) are rare, Euler angles provide an acceptable representation.

1.3.2 Rotation Matrices (SO(3))

A rotation matrix $R \in SO(3)$ is a 3×3 orthonormal matrix that transforms vectors from one coordinate frame to another. The notation $SO(3)$ stands for "Special Orthogonal group in 3 dimensions," representing all possible rotations in three-dimensional space.

A rotation matrix can be constructed from Euler angles using the product of elementary rotation matrices [3]

$$\boldsymbol{R(\varphi, \theta, \psi) = Rz(\psi) \cdot Ry(\theta) \cdot Rx(\varphi)}$$

Where Rx, Ry, Rz represent rotations around the X, Y, and Z axes respectively. The resulting 3×3 matrix contains nine elements, though only three are independent due to orthonormality constraints.

One drawback is redundancy: storing nine numbers to represent three degrees of freedom is inefficient Furthermore, they require periodic orthonormalization to correct numerical drift; common methods include Gram-Schmidt or singular value decomposition (SVD) [4].

The Mahony Passive Complementary Filter (Chapter 2) operates directly on rotation matrices, leveraging the geometric properties of $SO(3)$ for error correction. The filter computes a cross-product-based correction term that exploits the manifold structure of rotation matrices, providing superior performance compared to Euler angle approaches. To maintain orthonormality, the Mahony filter implementation applies Gram-Schmidt orthonormalization after each prediction step, ensuring the rotation matrix remains valid throughout the flight.

1.3.3 Quaternions

A quaternion q is a four-parameter representation of rotation, consisting of one scalar and one vector component:

$q = [q_0, q_1, q_2, q_3]^T = [q_0, q_v]^T$

Where:

q_0 : Scalar part

$q_v = [q_1, q_2, q_3]^T$: Vector part

Like rotation matrices, quaternions require normalization after numerical integration to maintain the unit constraint. However, this is computationally cheaper than matrix orthonormalization (single division versus full matrix projection).

Extended Kalman Filter (EKF) uses quaternions for the state representation, with a 7-dimensional state vector that includes the quaternion and gyroscope bias estimates.

1.3.4 Sensor Fusion Fundamentals

Sensor fusion combines measurements from multiple sensors to produce estimates superior to any single sensor. For attitude estimation, gyroscopes and accelerometers exhibit complementary error characteristics: gyroscopes provide smooth, high-frequency measurements but suffer from integration drift, while accelerometers offer a long-term gravity reference but are noisy and corrupted by linear acceleration.

The fundamental challenge is designing algorithms that:

1. Trust gyroscope measurements for short-term dynamics (responsiveness)
2. Use accelerometer measurements for long-term drift correction (stability)
3. Reject accelerometer corruption during vehicle acceleration

Fusion strategies range from simple weighted averaging (complementary filters) to nonlinear filters operating on manifolds (Mahony filter on SO(3)) to probabilistic state estimation (Extended Kalman Filter). Performance is evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) against ground truth measurements.

1.4 Neural Networks for Time-Series Predictions

While classical filtering approaches (detailed in Chapter 2) rely on explicit mathematical models of sensor behavior, modern deep learning offers a data-driven paradigm. This section establishes the theoretical foundation for the LSTM-based predictor developed in Chapter 3, moving from basic sequential learning to the attention mechanisms that enable robust multi-step ahead prediction.

1.4.1 Time-Series Data and Sequential Learning

Attitude estimation is fundamentally a time-series problem. IMU measurements arrive sequentially at fixed intervals Ts, and the vehicle's state at time t is highly dependent on its state at t-1. Traditional feedforward networks process each sample independently, ignoring this temporal structure. For a UAV, where dynamics are continuous, the model must understand these dependencies to distinguish between sensor noise and actual motion.

1.4.2 Recurrent Neural Networks (RNN)

RNNs address the limitations of feedforward networks by maintaining a hidden state h_t that acts as a "memory" of previous inputs. At each timestep t, the RNN processes the current input x_t and the previous hidden state h_{t-1}. While theoretically capable of learning long sequences, standard RNNs suffer from the vanishing gradient problem, where information

from earlier timesteps is lost during training [6]. For attitude prediction requiring look-back windows of K >= 10 samples, this makes standard RNNs unreliable.

1.4.3 Long Short-Term Memory (LSTM)

The LSTM architecture, introduced by Hochreiter and Schmidhuber [7], solves the vanishing gradient problem through a complex gating mechanism. It maintains a cell state C_t that carries information across long intervals. The flow of information is controlled by three gates:

Forget Gate (f_t): Determines which historical data to discard
Input Gate (i_t): Updates the cell state with new information
Output Gate (o_t): Filter the cell state to produce the hidden state h_t

For IMU data, LSTMs can learn to ignore high-frequency vibration noise (via the forget gate) while focusing on the persistent gravity vector (via the input gate).

1.4.4 Attention Mechanisms

Standard LSTMs use only the final hidden state h_k to generate predictions, which can create an information bottleneck. To improve prediction accuracy over a 10-step horizon (n=10) this work incorporates an **Attention Mechanism**. This layer calculates a score for every hidden state in the look-back window, allowing the model to "attend" to specific past events (e.g., a sudden change in ω_gyro) critical for predicting the future state, The final context vector c is a weighted sum:

$$c = \sum_{i=1}^{T} a_i h_i$$

where a_i represents the learned importance of the $t - th$ timestep. The weights a_i are computed by passing each hidden state through a scoring function followed by a softmax normalization, ensuring they sum to one. This allows the network to dynamically focus on different parts of the input sequence depending on the prediction task.

1.4.5 Related Work

Recent literature has explored deep learning as an alternative or complement to classic filters for IMU-based attitude estimation. Liu et al. [10] demonstrated LSTM-based orientation estimation for UAVs, showing that recurrent networks can learn temporal patterns in IMU data. Weber et al. [13] proposed RIANN, a GRU-based network that takes raw 6-DOF IMU data and outputs quaternion orientation, consistently outperforming Mahony, Madgwick, and complementary filters across diverse motion types and sensor hardware without per-dataset tuning. Golroudbari and Sabour [14] developed hybrid CNN-LSTM architectures evaluated across seven public datasets totaling over 120 hours of IMU measurements, achieving state-of-the-art accuracy for real-time attitude estimation. Brossard et al. [15] took a different approach, training a convolutional network to denoise raw gyroscope signals which are then integrated through dead reckoning, achieving accuracy comparable to visual-inertial odometry systems. Cohen and Klein [11] similarly investigated hybrid models for sensor denoising.

However, a common limitation of all these approaches is that they focus exclusively on **single-step estimation** of the current attitude state. None of them address the problem of **multi-step ahead prediction**, where the model must forecast future attitude angles over a prediction horizon of N timesteps. This distinction is critical: while accurate current-state estimation is sufficient for reactive controllers, advanced control strategies such as Model Predictive Control (MPC) require predictions of future states ($k + 1$ through $k + N$) to compute optimal control actions [12].

2 CHAPTER 2: Tilt Estimation Filters

This chapter presents the implementation, optimization, and comparative analysis of four classical sensor fusion algorithms for UAV attitude estimation. Building upon the theoretical foundations established in Chapter 1, we now examine how different mathematical approaches translate into practical filtering solutions, each offering distinct trade-offs between computational complexity, estimation accuracy, and robustness to sensor noise.

The four algorithms implemented in this work represent a progression of increasing mathematical sophistication. The Complementary Filter provides a baseline approach through simple weighted fusion of gyroscope and accelerometer measurements in Euler angle space. The Mahony Passive Complementary Filter advances this concept by operating on the $SO(3)$ manifold using rotation matrices, incorporating geometric error correction through proportional feedback. The Explicit Complementary Filter extends the Mahony approach by adding integral feedback for gyroscope bias estimation, providing adaptive correction for sensor drift. Finally, the Extended Kalman Filter (EKF) implements a full probabilistic state estimation framework using quaternion representation, offering optimal estimation under the assumptions of Gaussian noise and linear measurement models.

All implementations are developed in C++ using the Eigen library for efficient matrix operations, with careful attention to computational efficiency suitable for real-time embedded deployment. Each filter processes 3-axis gyroscope and accelerometer measurements from a 6-DOF IMU operating at 50 Hz sampling rate. The algorithms are validated against ground truth measurements from a dataset containing 1409 samples collected during quadcopter flight operations, covering a range of dynamic conditions from hover to aggressive maneuvers.

For each filter, this chapter presents the mathematical formulation, implementation details, hyperparameter tuning methodology, and performance evaluation using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics for both roll and pitch estimation. The analysis investigates how each algorithm handles the fundamental sensor fusion challenge: balancing short-term gyroscope accuracy against long-term accelerometer stability while rejecting accelerometer corruption during dynamic flight.

2.1 Quadcopter and Dataset

The quadcopter that used to gather data for both classic filter and LSTM, is built on an F450 FRAME and equipped with $1100kV$ brushless outrunner motors. It is paired with 8045 propellers and $30A$ electronic speed controllers, powered by a $2700mAh$ LiPo battery. Onboard avionics comprise a Navio2 autopilot interfaced with a Raspberry Pi 4 Model B. Inertial measurement is handled by dual IMUs — the MPU9250 and LSM9DS1 — each offering three-axis gyroscope, accelerometer, and magnetometer outputs.

Data were collected during real flight operations, in which a pilot maintained stable flight conditions while performing dynamic tracking of all three quadcopter orientation angles (roll, pitch, and yaw)

2.2 Complementary Filtering

The Complementary Filter is one of the most fundamental and widely used sensor fusion algorithms for attitude estimation in low-cost UAV applications due to its computational simplicity and effectiveness. It addresses the inherent limitations of individual inertial sensors by fusing their measurements in the frequency domain [8].

2.2.1 Algorithm Overview

The algorithm relies on the observation that gyroscope and accelerometer errors expose complementary spectral characteristics. Gyroscopes provide precise, low-noise measurements of angular velocity that are reliable in the high-frequency domain (short term) but suffer from low-frequency drift due to the integration of bias and noise over time. Conversely, accelerometers provide a stable absolute reference to the gravity vector that is reliable in the low-frequency domain (long term) but are subject to high-frequency noise and corruption from external linear accelerations (non-gravitational forces) during dynamic maneuvers.

The discrete-time implementation for the roll angle ϕ is given by:

$$\varphi(t) = \alpha \cdot \left(\phi(t-1) + \omega_{gyro} \cdot dt\right) + (1-\alpha) \cdot \phi_{accel}$$

2.2.2 Parameters

The parameter α control the trust that is placed on each sensor. It takes values in the range of $(0,1)$ and acts as a weighting factor that determines how much the filter relies on gyroscope integration versus accelerometer measurements. When α is close to 1, the filter places greater trust on the gyroscope term. On the other side when close to 0 the filter relies on the accelerometer.

To determine the optimal value, a grid search was performed over $\alpha \in [0.01, 0.99]$ with a step size of 0.01. For each candidate value, the Complementary Filter was executed on the full dataset and the RMSE of the ϕ estimation against ground truth was computed. The search identified $\alpha = 0.79$ as the value minimizing the roll RMSE.

2.2.3 Experimental results

The Complementary Filter was implemented with a gain coefficient $a = 0.79$ and tested against the ground truth dataset. The results for the Roll and Pitch estimation are presented below.

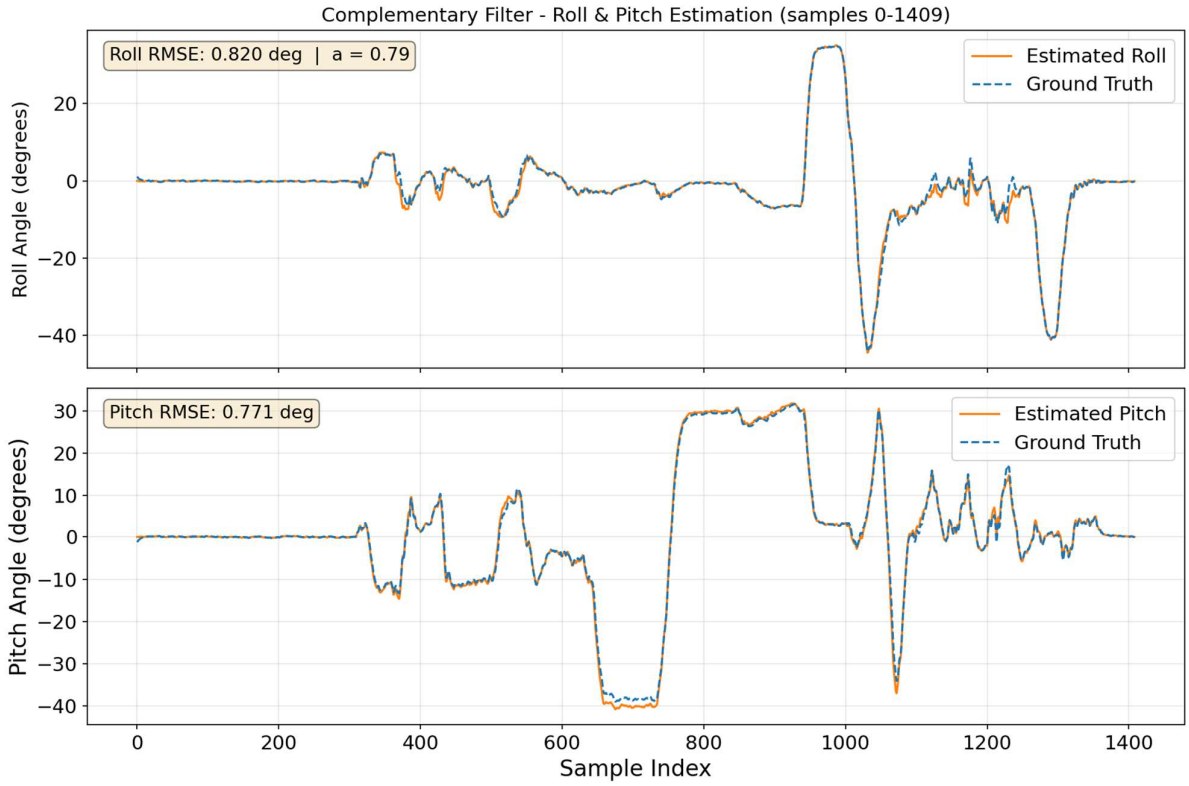

Figure 1 Complementary Filter Roll and Pitch Estimation

Figure 1 represents the global performance of the Complementary Filter over the entire flight duration (1409 samples). The estimated Roll and Pitch angles show a strong correlation with the ground truth data. The low RMSE values 0.820° for Roll and 0.771° for Pitch confirm that the filter successfully combines high-frequency gyroscope data with the low-frequency accelerometer data.

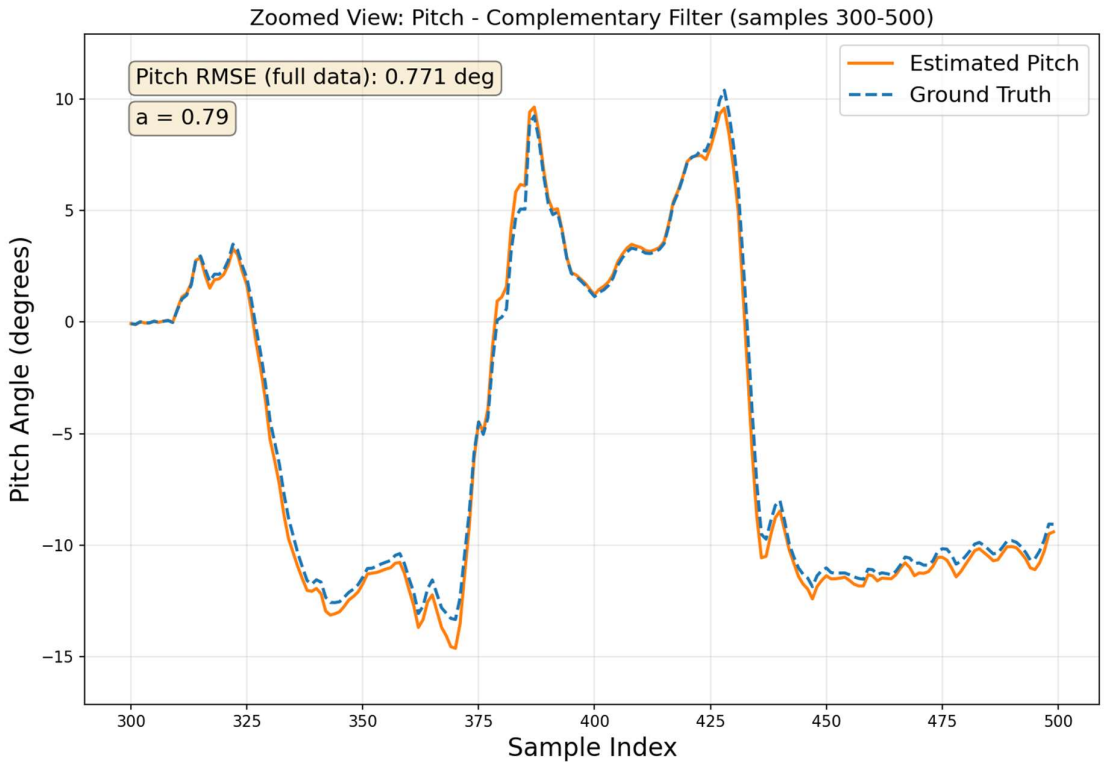

Figure 2 Complementary Filter Zoomed view (samples 300-500) Pitch Estimation

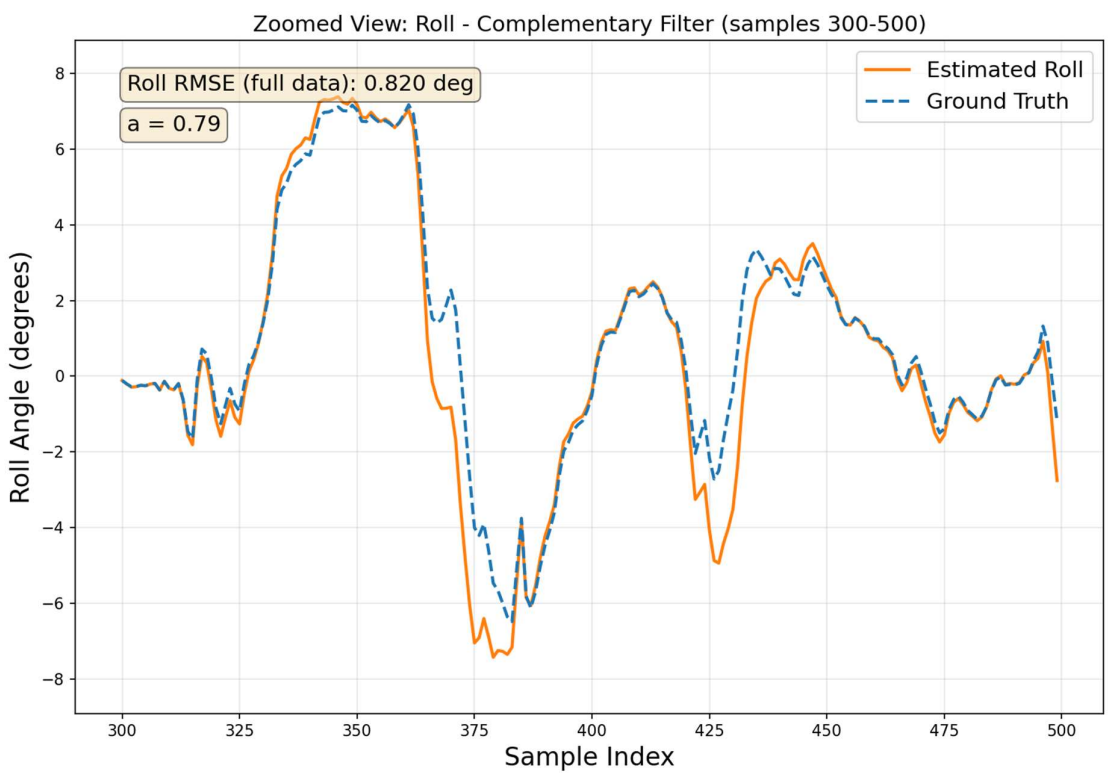

Figure 3 Complementary Filter Zoomed view (samples 300-500) Roll Estimation

Figure 2 provides a more detailed view of the Pitch estimation during a dynamic maneuvering phase (samples 300-500). It can be observed that the Pitch still follows close enough the truth data. Figure 3 illustrates the Roll estimation for the same interval. Generally, it performs almost as well as Pitch with a minor exception at around sample 375 which the filter fails to follow the truth data.

In conclusion, the Complementary Filter proves to be a robust and computationally efficient solution for general UAV attitude estimation, it successfully mitigates gyroscope drift and provides a smooth estimate suitable for flight control. The primary limitation is the reliance on a static α gain means it cannot adapt to various flight conditions that are encountered often in the real world.

2.3 Mahony Filter

The Mahony filter is a nonlinear complementary filter that operates directly on the Special Orthogonal group $SO(3)$, representing orientation using rotation matrices [4]. Unlike the standard Complementary Filter which operates on Euler angles, the Mahony filter avoids singularities (gimbal lock) and provides a more geometric approach to error correction.

The core principle of the Mahony filter is to estimate the rotation matrix $\hat{R}$ by comparing the measured direction of gravity (from the accelerometer) with the estimated direction of gravity. The error between these two vectors is used to generate a correction term that "steers" the gyroscope integration towards the true vertical.

The implementation done here is a "Passive" Mahony filter as described in [4].

2.3.1 Algorithm Overview

The Mahony filter maintains an estimated rotation matrix $\hat{R}$ that represents the current attitude. At each timestep, the filter compares the measured direction of gravity (derived from the accelerometer) with the direction predicted by the current estimate. The discrepancy between these two is used to generate a correction term that steers the gyroscope integration towards the true orientation.

A rotation matrix R_y is constructed from the accelerometer tilt angles, and the attitude error is computed as:

$$\tilde{R} = \hat{R}^T R_y$$

The correction vector is extracted from the anti-symmetric part of this error matrix:

$$\omega_{mes} = vec(\left(P_a(\tilde{R})\right), \quad P_a(\tilde{R}) = \frac{1}{2}(\tilde{R} - \tilde{R}^T)$$

The rotation matrix then updated using the corrected angular velocity:

$$\hat{\hat{R}} = \hat{R}\left[\omega_y + k_p \omega_{mes}\right]_\times$$

After each discrete integration step, $\hat{R}$ is projected back onto SO(3) via SVD orthonormalization to prevent numerical drift.

2.3.2 Parameters

The Mahony filter has a single tunable hyperparameter: the proportional gain k_p. This parameter controls the accelerometer-based correction applied to the gyroscope integration at each timestep.

When k_p is small (close to zero), the filter relies almost entirely on the gyroscope for attitude estimation. The gyroscope provides accurate short-term measurements but accumulates drift over time. When k_p is large, the filter applies aggressive corrections based on the accelerometer measurement. While this eliminates gyroscope drift, it make the estimate highly sensitive to accelerometer noise and non-gravitational acceleration that occur during maneuvers [4].

The optimal value of k_p must balance these two effects: it should be large enough to prevent gyroscope drift within the operating timeframe, yet small enough to maintain smooth estimates during dynamic flight. Unlike the Complementary Filter's a parameter which directly weights two angle estimates, k_p operates in the angular velocity domain, scaling the correction signal that is added to the gyroscope before integration.

To determine the optimal value, a grid search was performed over $k_p \in [1,100]$ with integer steps. For each candidate value, the Mahony filter was executed on the full dataset and the RMSE of both ϕ and θ was computed against ground truth. The optimization criterion was the average of the roll and pitch RMSE, ensuring that both angles are estimated well simultaneously. The search identified $k_p = 11$ as the optimal value.

2.3.3 Experimental Results

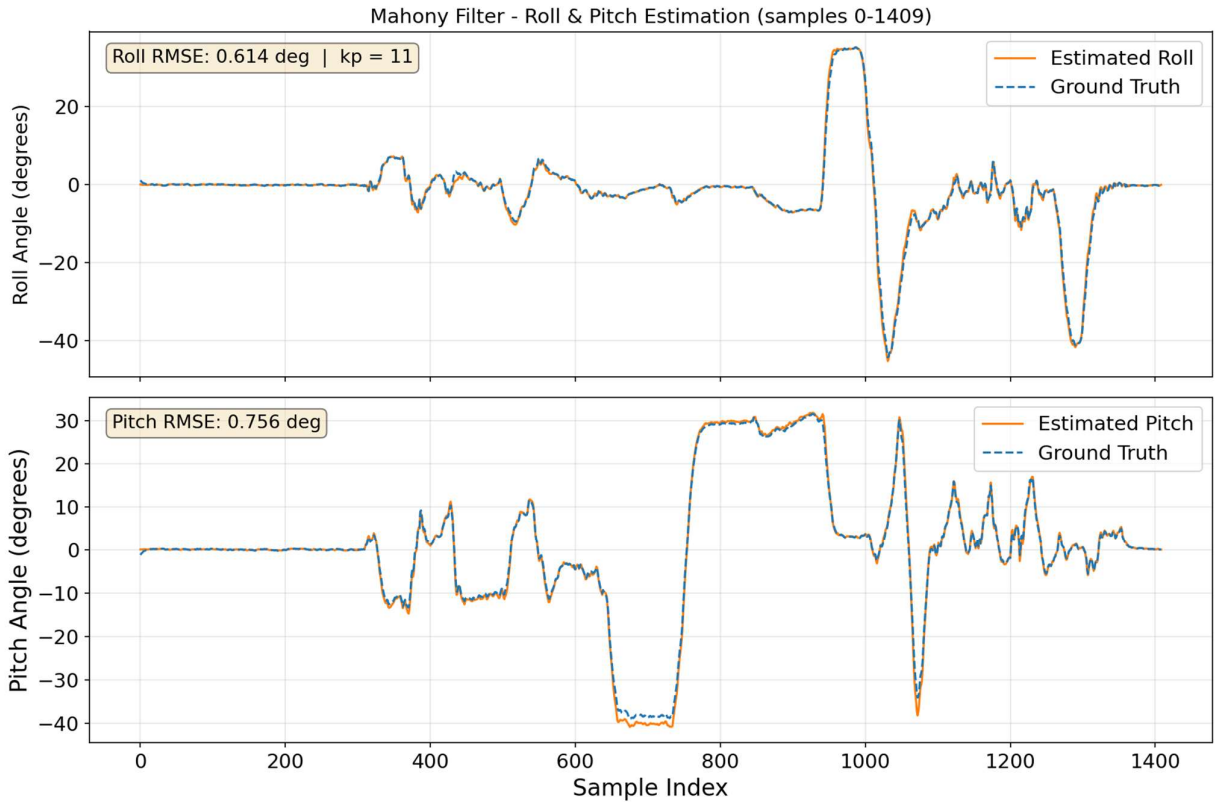

Figure 4 Mahony Filter Roll and Pitch Estimation

Figure 4 represents the global performance of the Mahony filter over the entire flight duration (1409) samples. The estimated ϕ and θ angles show a strong correlation with the ground truth data. The filter achieves an RMSE of 0.614 for ϕ and 0.756 for θ, representing a significant improvement over the Complementary Filter. The improvement is particularly notable for the roll angle ϕ, where the filter reduces the RMSE 25% approximately.

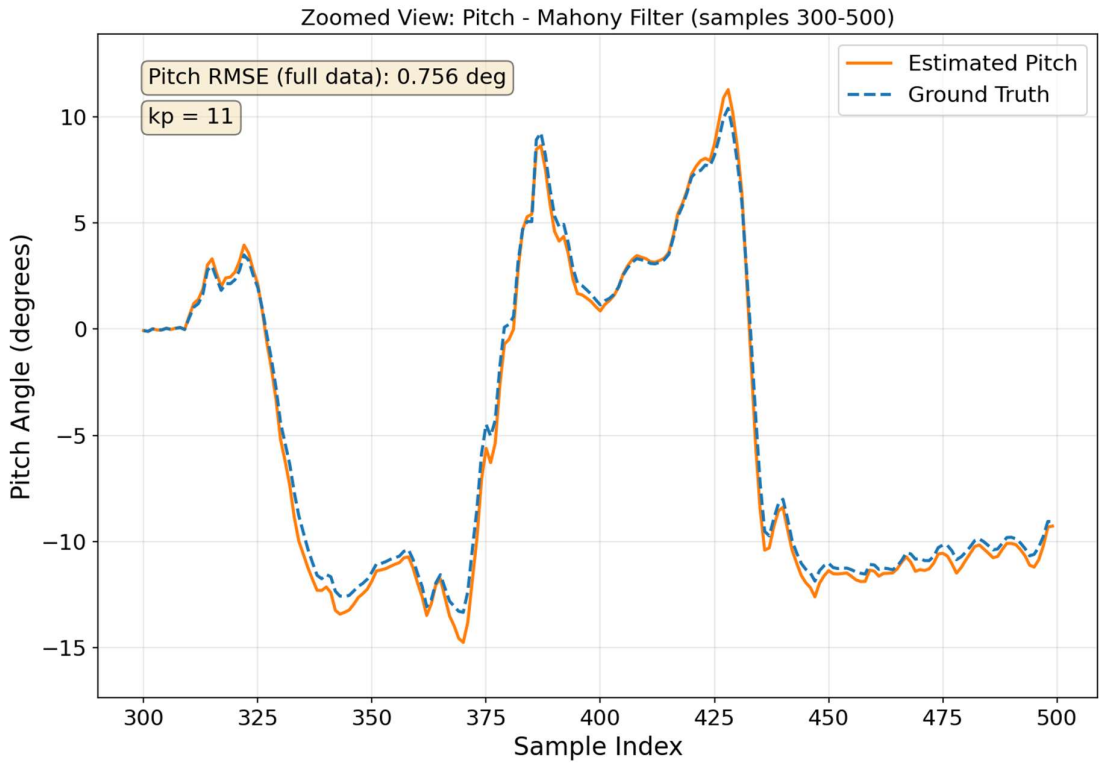

Figure 5 Mahony Filter Zoomed view (samples 300-500) Pitch Estimation

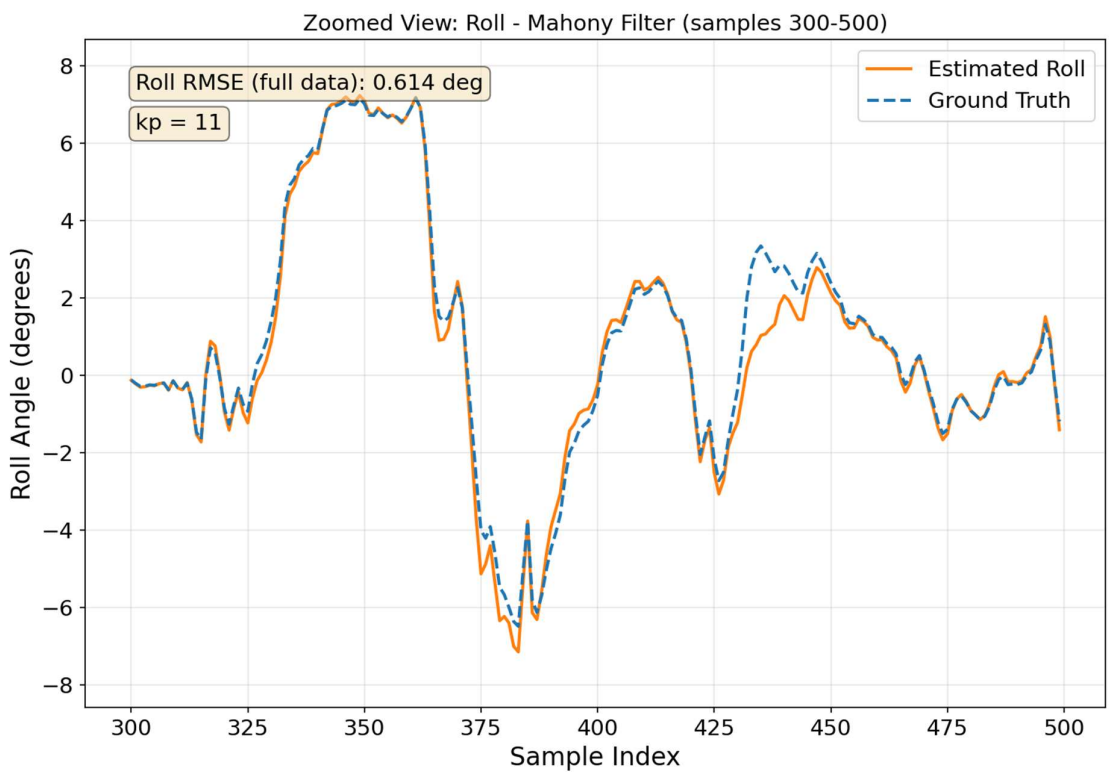

Figure 6 Mahony Filter Zoomed view (samples 300-500) Roll Estimation

Figure 5 and Figure 6 provide zoomed view of the attitude angles during a maneuvering phase (samples 300-500). Compared to the Complementary Filter, the Mahony filters the rapid angles changes more closely. The geometric error correction on $SO(3)$ allows the filter to handle large rotation more gracefully than the Euler angle-based approach of the Complementary Filter.

In conclusion, the Mahony Passive Complementary Filter demonstrates improved estimation accuracy over the basic Complementary Filter while maintaining computational efficiency suitable for real-time applications. Operating on $SO(3)$ with rotation matrices provides inherent robustness against gimbal lock and allows geometric error correction. The single hyperparameter k_p makes tuning straightforward, and the SVD-based orthonormalization ensures numerical stability during prolonged operation. However, the proportional-only correction means that persistent biases in the gyroscope may not be fully eliminated over very long time periods — a limitation addressed by the Explicit Complementary Filter in the following section.

2.4 Explicit Complementary Filtering

The Explicit Complementary Filter extends the Mahony approach by introducing two modifications: it operates directly on sensor measurements without requiring intermediate attitude reconstruction, and it incorporates integral feedback for online gyroscope bias estimation. Both features originate from Section V of Mahony et al. [4], where the filter is derived as an "explicit" complement to the passive formulation presented in the previous section.

2.4.1 Algorithm Overview

The filter maintains two state variables: an estimated rotation matrix $\hat{R}$ in $SO(3)$ and an estimated gyroscope bias vector $\hat{b}$ in R^3. At each timestep, the correction term is computed as the cross product between the measured and estimated gravity directions:

$\omega_{mes} = u \times \hat{u}, \hat{u} = \hat{R}^T u_0$

where $u_o = [0,0,1]^T$ is the gravity reference direction in the internal frame, u is the normalized accelerometer reading and $\hat{u}$ is the filter's predicted gravity direction in the body frame. The cross product produces a correction vector that is zero when the two directions are aligned, and grows proportionally to the angular error between them.

The two remaining core filter equations are:

$$\dot{\hat{R}} = \hat{R}\left[\omega^y - \hat{b} + k_p \omega_{mes}\right]_\times$$

$$\dot{\hat{b}} = -k_i \omega_{mes}$$

The first equation propagates the rotation matrix using the gyroscope measurement ω^y, corrected for the current bias estimate $\hat{b}$ and augmented with proportional feedback scaled by k_p. The second equation drives the bias estimate toward the true gyroscope bias over time through integral feedback scaled by k_i. After each discrete integration step, $\hat{R}$ is projected back onto $SO(3)$ via SVD orthonormalization to prevent numerical drift.

2.4.2 Parameters

The filter has two tunable hyperparameters: the proportional gain k_p and the integral gain k_i.

The proportional gain controls the correction applied to the gyroscope integration at each timestep, serving the same role as it in the passive Mahony filter. When k_p is small the filter relies primary on the gyroscope integration. On the other side a large number implies aggressive correction to reduce drift but increase sensitivity to accelerometer noise.

The integral gain k_i dictates the adaptation speed of the bias estimator. While an aggressive (larger) value ensures rapid convergence toward the true gyroscope bias, it risks inducing instability and oscillatory behavior in the attitude estimate. Conversely, a conservative (smaller) value prioritizes estimation stability and noise rejection.

To determine the optimal values, a two-dimensional grid search was performed over k_p in [1.0, 15.0] with a step size of 0.5 and k_i in [0.05, 1.0] with a step size of 0.05, yielding 580 parameter combinations. For each pair, the filter was executed on the full dataset and the average RMSE of phi and theta was computed against ground truth. The search identified k_p = 11 and k_i = 0.05 as the optimal combination.

2.4.3 Experimental Results

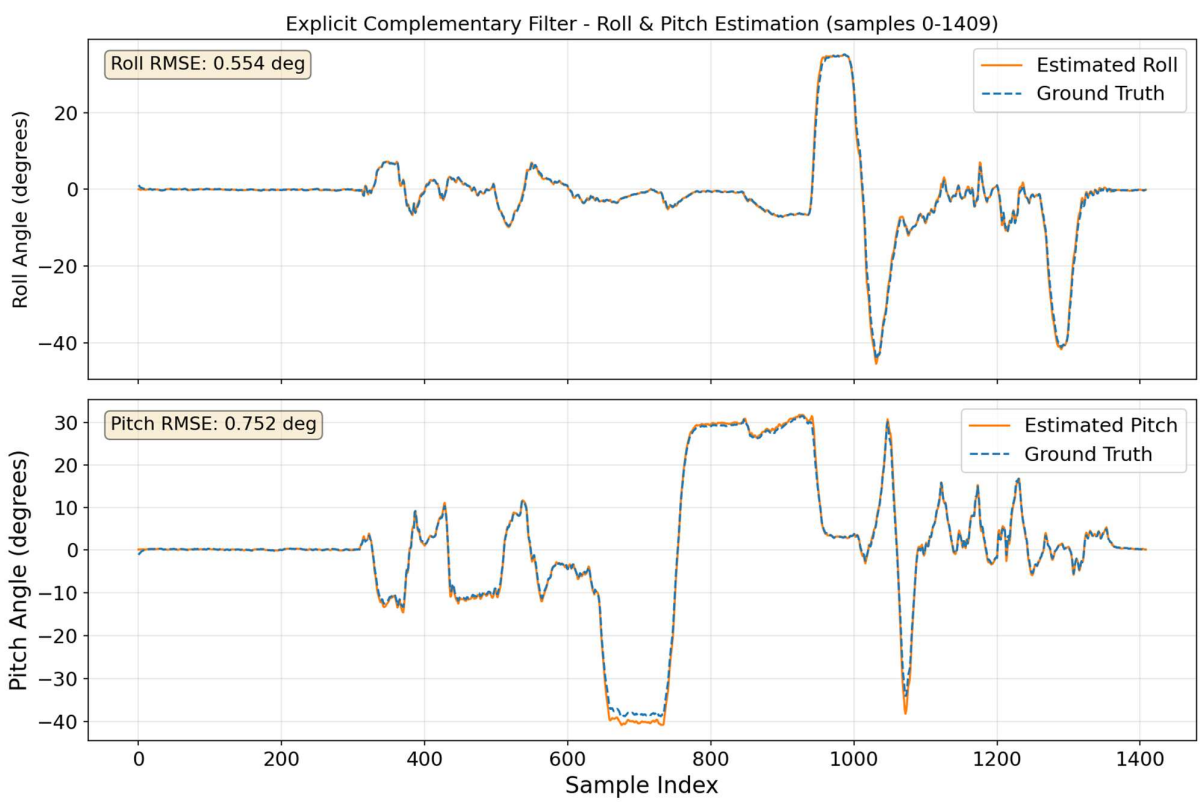

Figure 7 Explicit Complementary Filter Roll and Pitch Estimation

Figure 7 represents the global performance of the Explicit Complementary Filter over the entire flight duration. The filter achieves an RMSE of 0.554 degrees for ϕ and 0.752 degrees for θ. Compared to the "Passive" Mahony filter this one reduces the roll RMSE by an approximately 10% while maintaining comparable pitch performance.

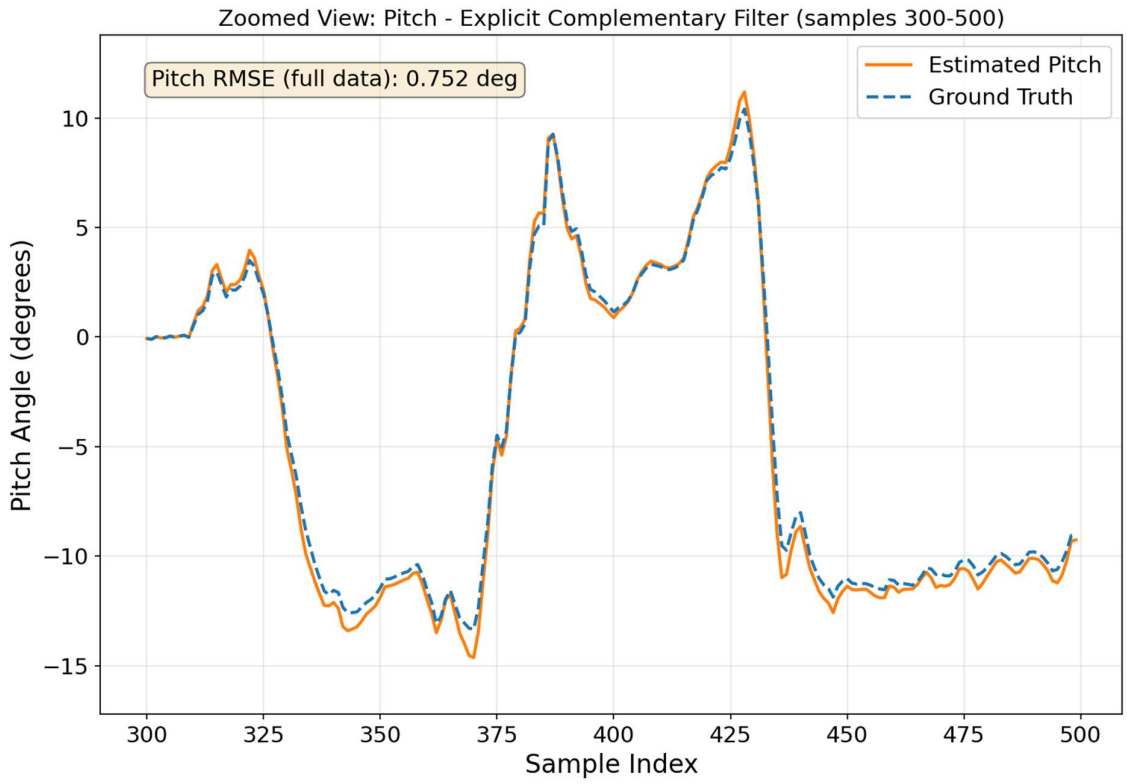

Figure 8 Explicit Complementary Zoomed view (samples 300-500) Pitch Estimation

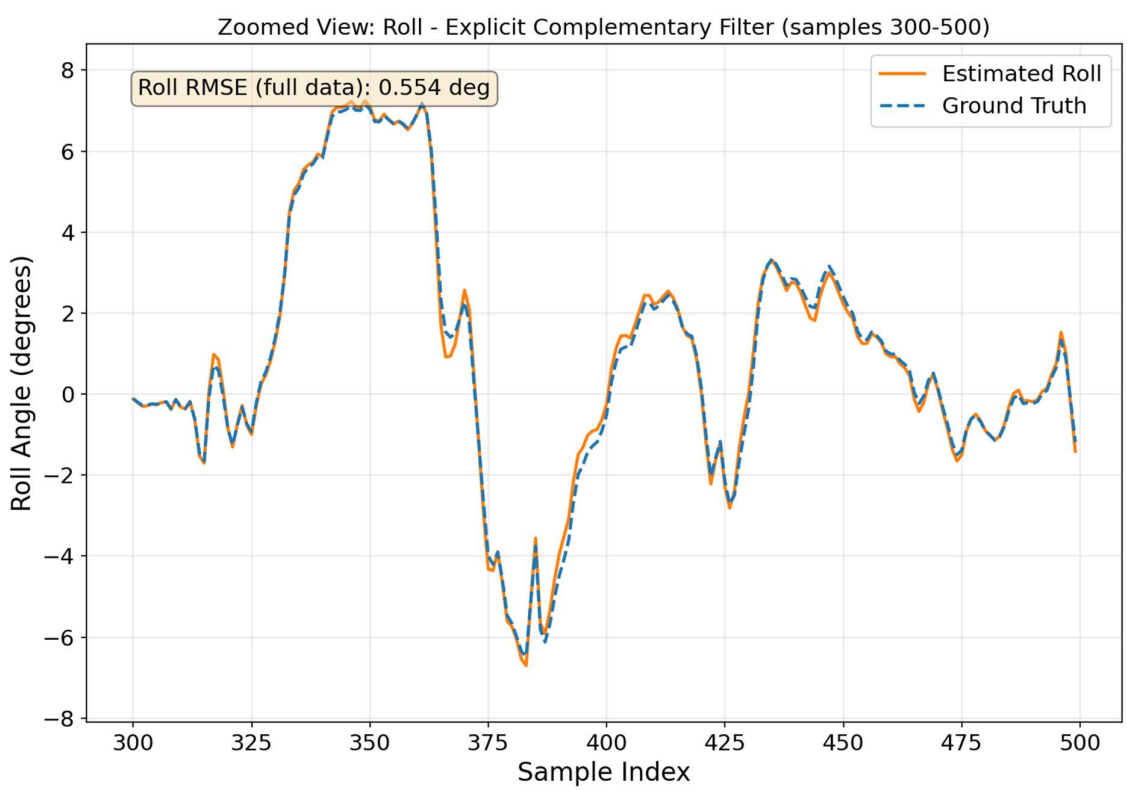

Figure 9 Explicit Complementary Zoomed view (samples 300-500) Roll Estimation

Figure 8 and Figure 9 provide a zoomed view of the attitude angles during a dynamic maneuvering phase. The estimates track the ground truth closely, with performance visually similar to the Mahony Filter. The improvement is most apparent in the roll angle, where the bias estimation mechanism provides a small but consistent reduction in tracking error.

In conclusion, the Explicit Complementary Filter demonstrates a modest improvement over the Passive Mahony Filter, primarily in roll estimation accuracy. Both filters share the same proportional correction mechanism operating on $SO(3)$, the same SVD-based orthonormalization, and the same optimal k_p value. The additional integral feedback for bias estimation provides a theoretical advantage that would become more pronounced in scenarios with significant gyroscope bias drift, which is not the case with the dataset. On the present dataset, where the gyroscope is well-calibrated, the improvement is limited to approximately 10% in roll RMSE. The filter's key architectural contribution, operating directly on vector measurements rather than requiring attitude reconstruction, reduces computational cost and eliminates potential numerical issues associated with the trigonometric extraction step, making it particularly suitable for resource-constrained embedded platforms.

2.5 Extended Kalman Filtering (EKF)

The Extended Kalman Filter is a probabilistic state estimation framework that extends the classical Kalman filter to nonlinear systems through the first-order linearization [16]. Unlike the complementary filters presented in the previous sections, the EKF maintains an explicit uncertainty estimate in the form of a covariance matrix, and fuses sensor measurements in a statistically optimal manner under the assumption of Gaussian noise.

The EKF represents attitude using a unit quaternion rather than a rotation matrix. Quaternions avoid the singularities inherent to Euler angle representations (gimbal lock) and are computationally more efficient than rotation matrices for integration [17]. The state vector is:
$x = \left[q_0, q_1, q_2, q_3, b_x, b_y, b_z\right]^T \epsilon\, R^7$

The $q = [q_0, q_1, q_2, q_3]^T$ is the unit quaternion encoding the current attitude and $b = \left[b_x b_y b_Z\right]^T$ is the estimated gyroscope bias. The filter alternates between a prediction step driven by the gyroscope and an update step driven by the accelerometer.

2.5.1 Algorithm Overview

Prediction step: The quaternion is propagated forward using first-order quaternion kinematics. The bias-corrected angular velocity is $w = \omega_y - b$ and the quaternion update is $q_{k+1} = q_k + \frac{\Delta t}{2}\Xi(w)q_k$ where $\Xi(w)$ is the quaternion derivative matrix constructed from the bias corrected angular velocity w. The covariance is propagated using the linearized state transition Jacobian F:
$P_{k+1} = F\, P_k F^T + Q$

Update step: The accelerometer provides a measurement of the gravity direction in the body frame. The expected measurement is computed by rotating the inertial gravity vector through the current quaternion estimate:

$h = R(q)^T g, \quad g = [0,0,1]^T$

The innovation $y = a_{norm} - h$, where a_{norm} is the normalized accelerometer reading, represents the discrepancy between the measured and predicted gravity directions. The Kalman gain and state update are then:

$K = PH^T(HPH^T + R)^{-1}$

$x \leftarrow x + Ky, \;\; P \leftarrow (I - KH)P$

2.5.2 Parameters

The EKF is governed by two noise covariance matrices: the process noise Q $(7x7)$ and the measurement noise R $(3x3)$[18]. These matrices encode the designer's trust in the process model versus the sensor measurements.

The process noise Q determines how much uncertainty is injected at each prediction step. A larger Q causes the filter to distrust its own propagation and rely more heavily on measurements, while a smaller Q makes the filter more confident in the gyroscope integration. Separate values are assigned to the quaternion components and the bias states, since these evolve at different timescales: the quaternion changes rapidly with motion, while the bias drifts slowly.

The measurement noise R encodes the expected noise level of the accelerometer. A larger R makes the filter discount accelerometer corrections, which is appropriate during dynamic maneuvers where non-gravitational accelerations contaminate the measurement. A smaller R increases responsiveness to accelerometer corrections at the cost of noise sensitivity.

For this implementation, the values were set to $Q_q = 0.001 * I_4$ (quaternion process noise), $Q_b = 0.0001 * I_3$ (bias process noise), and $R = 0.1 * I_3$ (accelerometer measurement noise). These values are consistent with standard practice for quaternion-based EKF attitude estimation [18].

2.5.3 Experimental Results

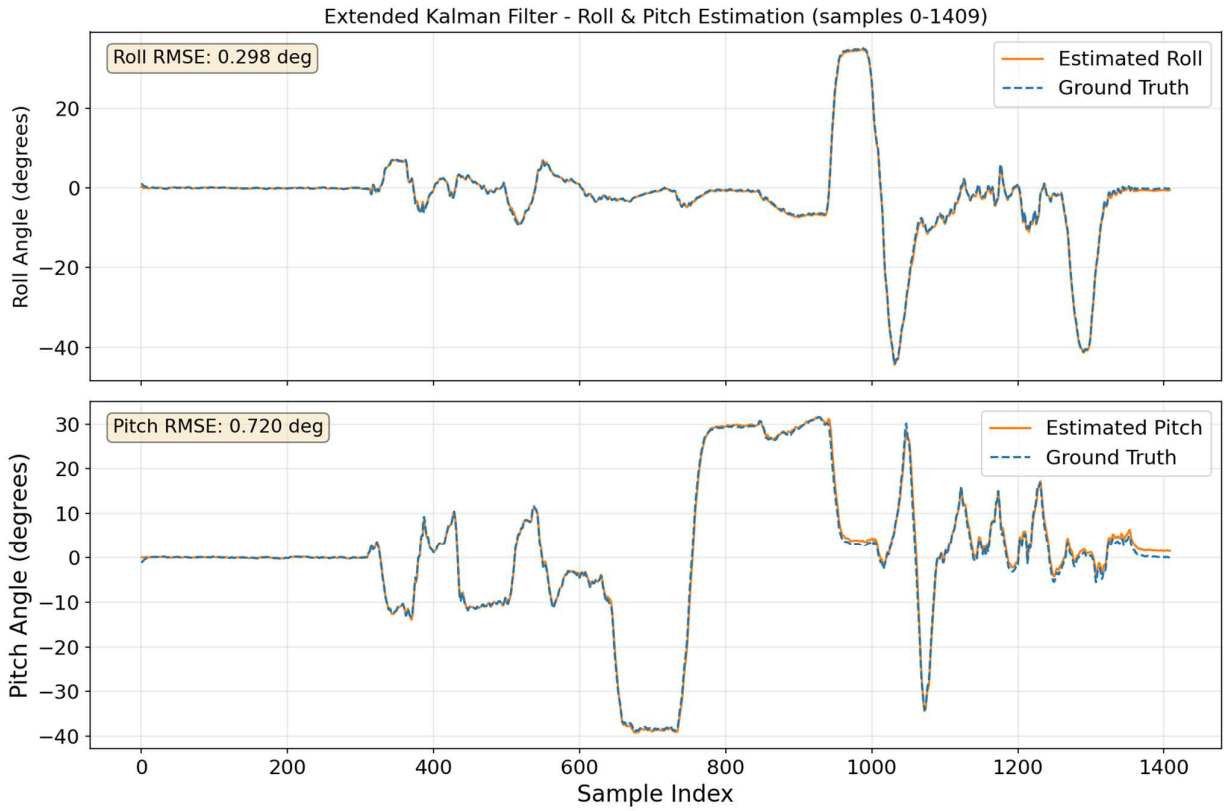

Figure 10 EKF Roll and Pitch Estimation

Figure 10 represents the global performance of the EKF over the entire flight duration. The filter achieves an RMSE of 0.298 degrees for ϕ and 0.720 for θ. The roll estimation represents that best performance among all four filters implemented in this chapter, with a 51% reduction in roll RMSE compared to the basic Complementary Filter and a 52% reduction compared to Mahony Filter.

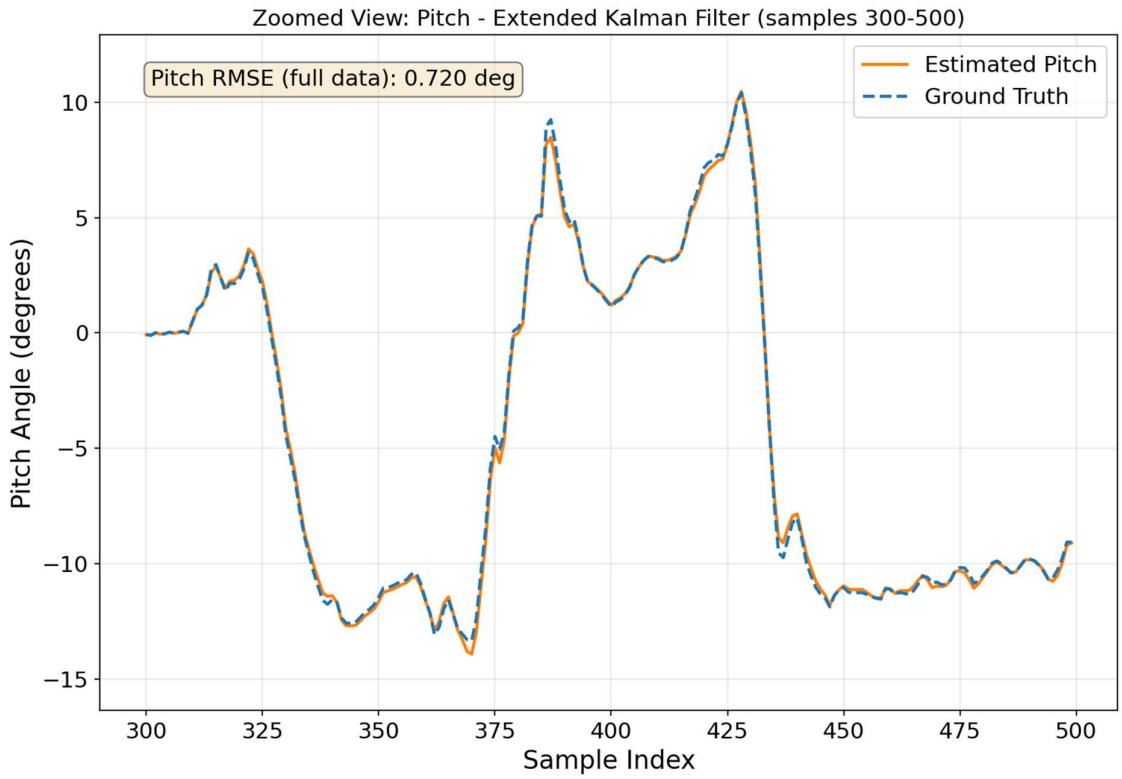

Figure 11 EKF Zoomed view (samples 300-500) Roll Estimation

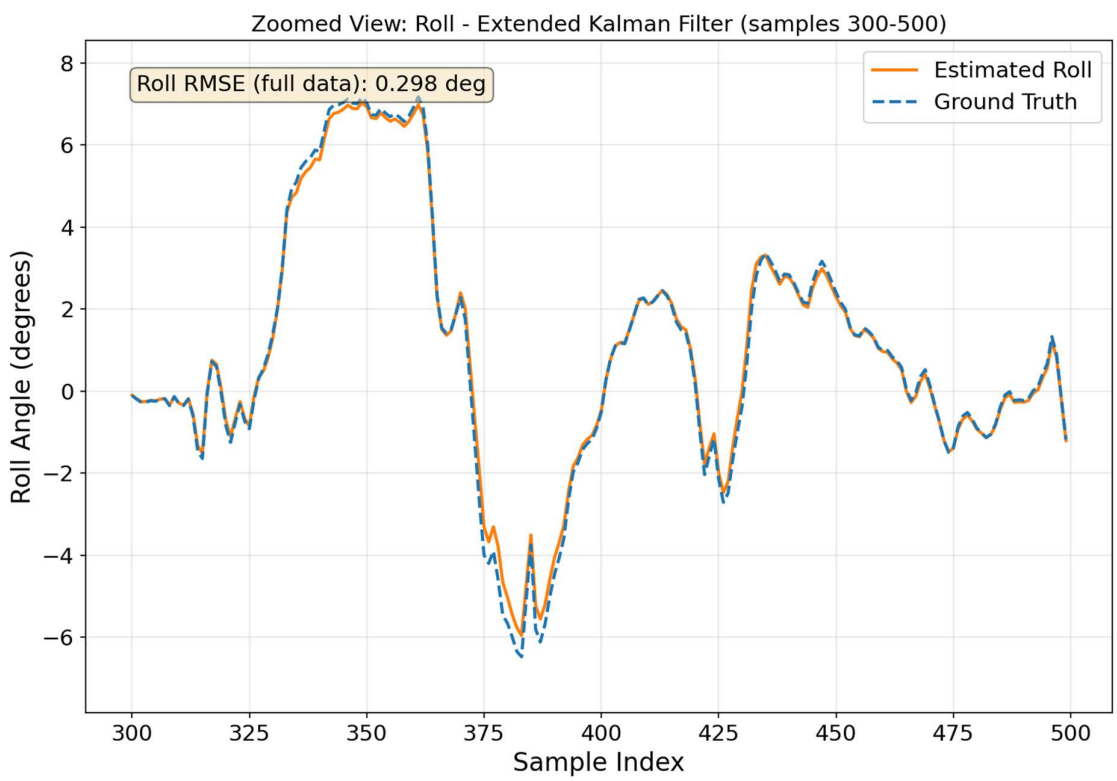

Figure 12 EKF Zoomed view (samples 300-500) Pitch Estimation

Figure 11 and Figure 12 provide a zoomed view during the dynamic maneuvering phase (samples 300-500). The EKF tracks the ground truth more closely than the complementary filters, particularly in the roll axis where the bias estimation and probabilistic weighting provide the most benefit.

In conclusion, the EKF achieves the highest roll estimation accuracy of the four filters, demonstrating the advantage of probabilistic state estimation with explicit uncertainty tracking. However, this performance comes at a higher computational cost: at each timestep the filter maintains and updates a 7x7 covariance matrix and computes a 3x7 Kalman gain, making it more demanding than the complementary filter approaches. The tuning of the Q and R matrices also requires more domain knowledge than the single-gain tuning of the Mahony and Explicit CF filters.

2.6 Limitations and Conclusions

It is important to acknowledge a methodological limitation regarding the classical filter evaluation: the hyperparameters (such as, α, k_p, and k_i) were optimized via grid search over the entire available flight dataset rather than a strictly held-out validation set. While this approach inherently carries a risk of overfitting to this specific flight profile, it was deemed an acceptable trade-off for this specific phase of the research. The primary objective of implementing these classical filters was not to evaluate their generalized standalone performance, but rather to extract the highest possible quality of baseline attitude estimates from this specific dataset. These optimized estimates act as the foundational training labels for the deep learning predictive pipeline developed in Chapter 3.

Filter Algorithm	**Roll RMSE (degrees)**	**Pitch RMSE (degrees)**	**Key Characteristics**
Complementary	0.820	0.771	Computationally lightweight; Euler-based; static trust gain.
Mahony (Passive)	0.614	0.756	Operates on $SO(3)$. Geometric error correction; single gain (k_p)
Explicit CF	0.554	0.752	Direct vector measurements; adds integral gain (k_i) for bias.

Extended Kalman Filter (EKF)	0.298	0.720	Full probabilistic state estimation. Optimal Gaussian fusion.

Table 1 Summary of Classical Filter Performance

As demonstrated in Table 2.1, the Extended Kalman Filter achieves the superior estimation accuracy across both axes, significantly outperforming the simpler complementary approaches, particularly in roll estimation.

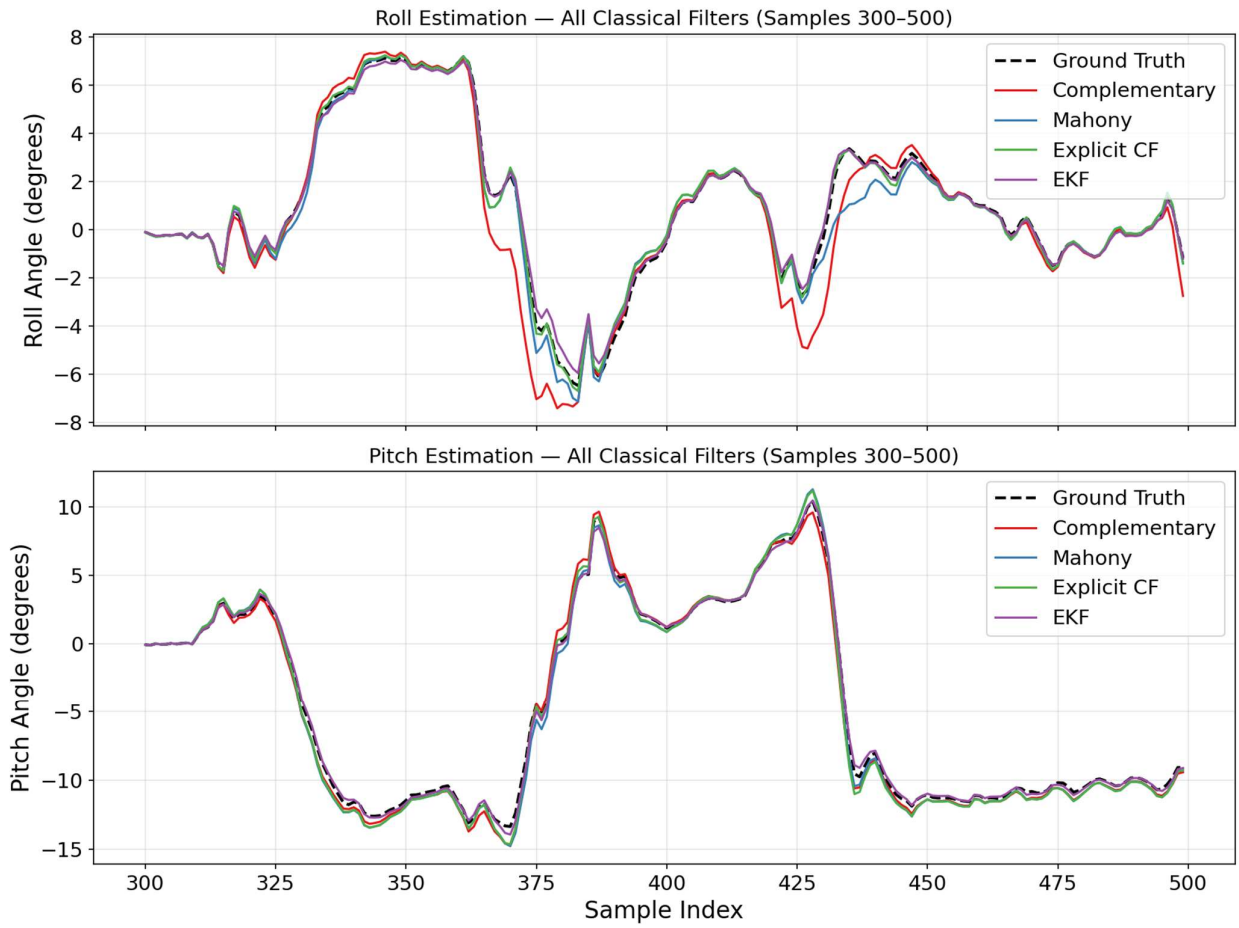

Figure 13 Filters comparison in specific window

To visually consolidate the comparative analysis, Figure 13 presents the estimation performance of all four classical filters plotted simultaneously against the ground truth during a dynamic maneuvering phase (samples 300-500).

3 Chapter 3: Deep learning approach

This chapter introduces an attention-based Long Short-Term Memory (LSTM) network designed for the multi-step ahead prediction of Unmanned Aerial Vehicle (UAV) attitude

angles. While Chapter 2 demonstrated that classical tilt estimation filters can accurately resolve the current orientation of a UAV, these algorithms are fundamentally reactive; they process sensor measurements as they arrive to produce an estimate of the present state. Consequently, such algorithms lack the inherent capacity to anticipate how the vehicle's attitude will evolve over a future temporal horizon.

This chapter addresses a significantly more demanding problem: given a look-back window of recent observations, the objective is to predict the attitude angles (ϕ, θ, ψ) over a future horizon of N timesteps. This predictive capability is essential for advanced flight control strategies, such as Model Predictive Control (MPC), and its realization through a specialized deep learning architecture represents the core innovation of this thesis.

3.1 Motivation: From Reactive Estimation to Predictive Control

The four tilt estimation filter developed in Chapter 2 share a common principle. At each discrete timestep k, these algorithms fuse gyroscope and accelerometer measurements to produce an estimate of the current attitude angles. The EKF which achieved the superior performance with a roll RMSE of 0.298 and pitch RMSE of 0.720, represents the upper bound of what reactive estimation can achieve on the provided dataset. Despite their mathematical formulation, ranging from simple weighted averaging to full probabilistic state estimation with covariance tracking, all four filters address a singular fundamental question: what is the vehicle's orientation at the current instant?
While this estimation is sufficient for basic flight stabilization, where a controller reacts to deviations from a desired attitude by applying corrective torques, a purely reactive control loop is constrained. The controller can only respond to error states after they happen, introducing a response latency that downgrades performance during aggressive maneuvers, rapid trajectory changes, or external disturbances such as wind gusts.

Modern advanced flight control strategies deal with this limitation through prediction. Model Predictive Control (MPC), is one of the most widely adopted frameworks for autonomous UAV flight [19] and it operates by solving an optimization problem over a finite future horizon at each control step [12]. Rather reacting to the current error, MPC computes a sequence of optimal control actions by evaluating predicted future trajectories. The controller then applies only the first of this sequence, re-solves the optimization at the subsequent timestep with updated measurements and repeats the process. This approach enables the controller to anticipate the consequences of its actions and proactively react to upcoming disturbances.

The critical prerequisite for MPC is a prediction model capable of predicting the system state and more specifically, the attitude angle over the planning horizon. Given the current timestep k, the controller requires accurate predictions of (ϕ, θ, ψ) at timesteps $k+1, k+2, \dots k+N$. The accuracy of these predictions directly affect the correctness of the control actions.

Classical physics-based prediction models exist for UAV dynamics, but they require accurate knowledge of the vehicle's physical parameters (mass, moments of inertia, and aerodynamic coefficients) and are very sensitive to modeling inaccuracies [20]. These limitations create a

need for a different approach. Deep learning models, and recurrent neural networks in particular, can learn the temporal dynamics of the system directly from the flight data without requiring explicit physical models.

The theoretical foundations for this approach were established in the Section 1.4, where recurrent neural networks, LSTM architecture, and attention mechanisms were introduced. While the majority of existing research focuses on single-step state estimation, the domain of multi-step ahead prediction for attitude control remains less explored.

This thesis tried to bridge this gap. The proposed architecture is designed to process a look-back window of K timesteps, containing attitude estimates, gyroscope measurements, and control torques, to generate predictions of roll (ϕ), pitch (θ), and yaw (ψ) for the subsequent N timesteps. A key feature of this design is an addition of an attention mechanism, which enables the network to dynamically weight the importance of past observations. This avoids the information bottleneck inherent in standard recurrent architectures that compress the entire input sequence into a single fixed-length state vector. The resulting implementation is a predictive model engineered to serve as a core component within an MPC framework.

3.2 Problem Formulation

This section formally defines the multi-step ahead prediction task, describes the input and output data representations, and presents the dataset and preprocessing pipeline used for training.

3.2.1 Prediction Task Definition

The objective is to learn a mapping f that given a sequence of K consecutive observations vectors, predicts the attitude angles over the next N timesteps. Formally, let $x(t) \epsilon R^d$ denote the observation vector at timestep k, where d is the number of input features. The prediction tasks are:

$$f(x(k), x(k+1), \dots, x(k+K-1) = (\hat{y}(k+K), \hat{y}(k+K+1), \dots\dots, \hat{y}(k+K+N-1))$$

where $\hat{y}(t) = \left[\hat{\phi}(t), \hat{\theta}(t), \hat{\psi}(t)\right]^T \epsilon\, R^3$ is the predicted attitude vector at future timestep k. The input windows spans timesteps k through $k+K-1$ (the look-back windows), and the output spans timesteps $k+K$ through $k+K+N-1$ (the prediction horizon).
Both K and N are configurable hyperparameters. At a sampling rate of 66.6 Hz ($T_s = 0.015s$) a look-back window of K timesteps covers $K\ X\ 0.015$ seconds of flight history and the prediction horizon of N timesteps extends $N\ X\ 0.015$ seconds into the future. The effect of different K and N configurations on prediction accuracy is evaluated experimentally in Section 3.5.

3.2.2 Input Feature Vector

Each observation vector $f(x) = k \in R^9$ is composed of three distinct feature groups that capture various details of the UAV's characteristics.

1. **Attitude Angle Estimates ($\boldsymbol{\phi}, \boldsymbol{\theta}, \boldsymbol{\psi}$)** – Columns [0-2]: These represent the roll, pitch, yaw angles in radians. In a deployed system these values will be provided by the EKF.

Integrating historical attitude estimates enables the model to internalize the temporal evolution of the vehicle's orientation, providing the necessary context for forecasting future states.

2. **Gyroscope Angular Velocities ($\boldsymbol{\omega_\phi, \omega_\theta, \omega_\psi}$)** – Columns [3-5]: These are the angular rates measured by the IMU's 3-axis gyroscope in rad/s. From a prediction perspective these angular velocities serve as robust indicators of near-term orientation changes.
3. **Control Torques ($\boldsymbol{\tau_\phi, \tau_\theta, \tau_\psi}$)** – Columns [6-8]: These represent the torques generated by the PID flight controller. The control torques are the causal drivers of attitude change, as they control the angular acceleration applied to the vehicle. Providing the network with direct access to these inputs allows it to distinguish between commanded maneuvers and external disturbances, which would otherwise have to be inferred indirectly.

The synthesis of these feature groups provides the network with a comprehensive representation of the vehicle's dynamics. The attitude angles describe the current state, the gyroscope rates describe the rate of change, and the control torques describe the forces initiating that change. Together, these features encode the position, velocity, and forcing terms of the attitude dynamics—analogous to the components of a second-order dynamical system

3.2.3 Output Representation

The network output is a matrix $\hat{Y} \in R^{N\times 3}$ containing the predicted angles for the N timesteps of the prediction horizon:

$$\hat{Y} = \begin{bmatrix} \hat{\phi}(k+K) & \hat{\theta}(k+K) & \hat{\psi}(k+K) \\ \hat{\phi}(k+K+1) & \hat{\theta}(k+K+1) & \hat{\psi}(k+K+1) \\ \vdots & \vdots & \vdots \\ \hat{\phi}(k+K+N-1) & \hat{\theta}(k+K+N-1) & \hat{\psi}(k+K+N-1) \end{bmatrix}$$

The output dimensionality depends on the prediction horizon $N\ X\ 3$. The architecture employs a **Direct Multi-Output** strategy, where the fully connected layer generates all values simultaneously in a single forward pass.

3.2.4 Dataset

The LSTM network is trained on a combined dataset constructed from seven separate flight recordings, a different strategy from the single recording data that was used in Chapter 2. The individual recordings were merged into a unified dataset containing 21,164 samples record at a sample frequency of 66.6 Hz ($T_s = 0.015s$), corresponding approximately to 317 seconds (~5.3 minutes) of total flight time. Each sample is a 9-dimensional vector containing the three feature groups described in Section 3.2.2.

The use of multiple flight data is an intentional choice. A model trained on single flight data has a risk of learning specific patterns rather than general attitude dynamics. By combining seven flights with various maneuvering profiles the dataset exposes the network to a wide range of operating conditions. Roll has a range from $13.3°$ to $-24.8°$, pitch from $20.1°$ to

$-14.7°$ and yaw from $22.4°$ to $-21.7°$. This diversity is critical for model in order generalize instead of memorizing.

3.3 Network Architecture

This section presents the architecture of the proposed attention-based LSTM network. The design follows a multi-stage pipeline: a recurrent layer encodes the temporal patterns in the input sequence, dropout layers provide regularization, an attention mechanism selects the most relevant information from the encoded sequence, and a fully connected layer maps the attended representation to the multi-step output. The complete architecture is summarized as:

```
Input(9) → LSTM(H) → Dropout(p) → Attention -> Dropout(p) →
FC(N X 3) → Reshape(N x 3)
```

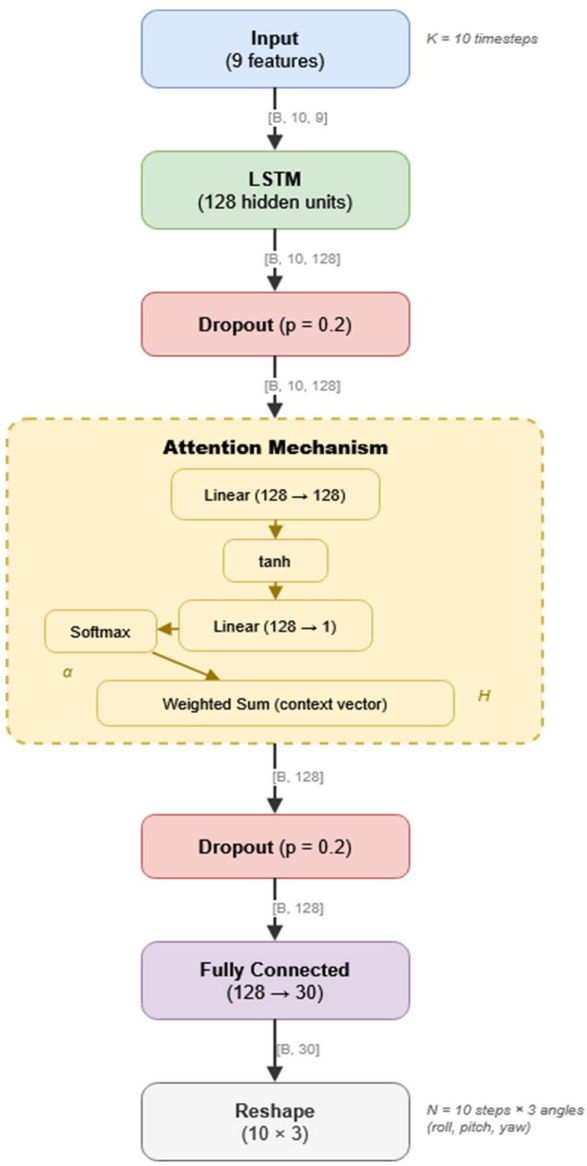

Figure 14 LSTM Attention Based Network Architecture for K=10

3.3.1 LSTM Layer

The first stage of the network is a single LSTM layer that processes the input sequence and produces a hidden state representation at each timestep. As introduced in Section 1.4.3, the LSTM maintains a cell state c_t and a hidden state h_t that are updated at each timestep through the forget input and output gates. This gating mechanism allows the network to selectively retain or discard information across timesteps, addressing the vanishing gradient problem that limits standard RNNS on sequences of this length.

The LSTM receives the input tensor $X \in R^{BxKxd}$, where B is the batch size, K is the look-back window length, and $d = 9$ is the number of input features. At each timestep k, the LSTM processes the feature vector $x_t \in R^9$ together with the previous hidden state h_{t-1} and cell state c_{t-1} producing and updated hidden state $h_t \in R^H$ where H is the hidden state dimensionality. The output of the LSTM layer is the full sequence of hidden states:

$$H = [h_1, h_2, \dots, h_K], h_t \in R^H$$

Crucially, unlike standard recurrent architectures that discard intermediate states and only utilize the final hidden state h_K for prediction, this design retains the entire sequence $\boldsymbol{H}$. By preserving the hidden state at every timestep, the network maintains a dense temporal context that is required for the subsequent attention mechanism.

3.3.2 Attention Mechanism

The attention mechanism is the key architectural component that distinguishes this network from standard LSTM predictors. Its purpose is to compute a weighted combination of all hidden states $h_1, \dots, h_K$, allowing the network to dynamically determine how much each timestep in the look-back window contributes to the final prediction. The implementation utilizes an additive scoring function to compute a context vector $f(c) \in R^H$. For each hidden state h_t, an alignment score s_t is calculated through a learned non-linear projection:

$$s_t = v^T \tanh(W_a \cdot h_t + b_a)$$

where $W_a \in R^{H \times H} and\ v \in R^H$ are the weight matrices and vectors, respectively. These scores are normalized via a softmax operation to produce the attention weights α_t:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{j=1}^{K} \exp(s_j)}$$

The final attended representation is computed as the weighted sum of the hidden states: $c = \sum_{t=1}^{K} \alpha_t \cdot h_t$. This allows the network to prioritize informative timesteps regardless of their position in the look-back window, which is critical for capturing the varying dynamics of UAV flight.

3.3.3 Dropout Regularization

Training a neural network on a dataset of finite samples has a risk of overfitting, as the model memorizes specifics patterns instead of generalizing. One of the most effective ways to handle this is to add regularization dropout [5] at two points in the forward pass:

1. **After the LSTM hidden states** - before the attention mechanism process them This prevents the attention layer from becoming very reliant on specific hidden units and encourages the LSTM to distribute information across the full hidden state.
2. **After the context vector –** before the fully connected output layer. This prevents the output layer from adapting to specific combinations of attended features

Both dropout layers use a rate of $p = 0.2$, meaning that during each training forward pass, 20% of the values in the affected tensors are randomly set to zero and the remaining values are scaled by $\frac{1}{1-p}$ to maintain the expected magnitude. During evaluation dropout is disabled and all units are active ensuring deterministic predictions.

3.3.4 Output Layer

The context vector $c \in R^H$ is passed through a single fully connected layer that maps it to $N\ X\ 3$ output values, which are then reshaped into the prediction matrix $\hat{Y} \in R^{N\ X\ 3}$. No activation function is applied, as this is a regression task requiring unconstrained output values. A single output layer is sufficient because the attention mechanism already provides a rich, nonlinearly-processed representation of the input; additional depth would increase complexity without proportional benefit for the low output dimensionality.

3.4 Training Methodology

This section describes the complete training pipeline: how training samples are constructed from the raw time series, how the data is split for evaluation, the preprocessing applied, and the optimization configuration used to train the network.

3.4.1 Sliding Windows

Training samples are constructed using a sliding window approach over the concatenated time series. For each index i (where $0 \leq i \leq S - K - N$ with $S = 21{,}164$, the total number of samples), a training pair X_i, Y_i is constructed:

Input:$X_i \in R^{K \times d}$- rows i through $i + K - 1$ of the dataset, containing K consecutive observation vectors with $d = 9$ features each.

Target:$Y_i \in R^{N \times 3}$ – rows $i + K$ through $i + K + N - 1$, containing only the attitude angle angles for the next N timesteps.

With $S = 21{,}164$, this yields $S - K - N + 1$ sliding windows. The exact number depends on the chosen K and N values. The resulting tensor shapes are:

- Input tensor $\boldsymbol{X}$:[windows, K, 9]
- Target tensor $\boldsymbol{Y}$: [windows, N, 3]

Because the dataset is formed by concatenating seven separate flights, a small number of windows will span the boundary between two flights. At these boundaries, the last samples of one flight are adjacent to the first samples of the next, potentially introducing discontinuities

in the input sequence. In practice, this affects at most 6 windows and has minor impact on training.

3.4.2 Data Standardization

Neural networks are sensitive to the scale of input features. The nine features in this dataset have different measurements. Attitude angles are in radians, angular velocities in rads/s and control torques in Nm, each of them has different characteristic magnitudes. To address this, z-score standardization [21] is applied to the entire dataset. Each feature column is transformed to zero mean and unit variance using the column-wise mean and standard deviation.

The standardization statistics are computed once and stored for use during inference. Since the network outputs predictions in the standardized space, all outputs are inverse-transformed back to radians before computing evaluation metrics. All results reported in Section 3.5 are in the original physical scale.

3.4.3 Data Splitting

To evaluate generalization performance, the sliding windows are split into three sets:

Set	Proportion	Purpose
Training	80%	To train the model
Validation	10%	Monitor training progress and overfitting
Test	10%	Performance evaluation

Table 2 Data split percentage

Before splitting, all windows are randomly shuffled. This is essential because the sliding windows are constructed sequentially from the concatenated flights. The test set would consist entirely of windows from the last flight, providing no assessment of the model's ability to generalize across different flight profiles. The shuffled split ensures that each set contains samples from all seven flights.

To monitor training progress, the validation loss is computed at five-epoch intervals with dropout disabled. This allows for real-time assessment of the model's generalization capabilities and assists in detecting potential overfitting. The test set, however, is reserved entirely for post-training evaluation to ensure that the final results presented in Section 3.5 reflect the model's performance on previously unseen data.

3.4.4 Loss & Optimization

The network is trained to minimize Mean Squared Error (MSE) between predicted and target attitude angles. MSE penalizes large error quadratically, encouraging the network to avoid

large deviations. The loss is computed on standardized values, ensuing all three angles contribute equally regardless of their original physical scales.

The model is trained using the Adam optimizer with the following configuration:

Parameter	Value
Loss function	MSE
Optimizer	Adam
Learning rate	0.001
Batch size	64
Epochs	500
Dropout rate	0.2

Table 3 Optimizer Configuration

At each epoch, the training sample indices are shuffled and processed in mini-batches of 64. For each batch, a forward pass computes the predictions and loss, gradients are computed via backpropagation, and the optimizer updates the model parameters.

3.5 Experimental Results and Evaluation

TODO Results

3.6 Discussion

The experimental results validate the proposed attention-based LSTM architecture as an effective approach for multi-step ahead UAV attitude prediction. This section discusses some key findings, the role of the attention mechanism, and the limitations of the current work.

3.6.1 The role of Attention

The comparison between attention and non-attention models

Conclusions

TODO Conclusions

Future Work

TODO Future Works

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